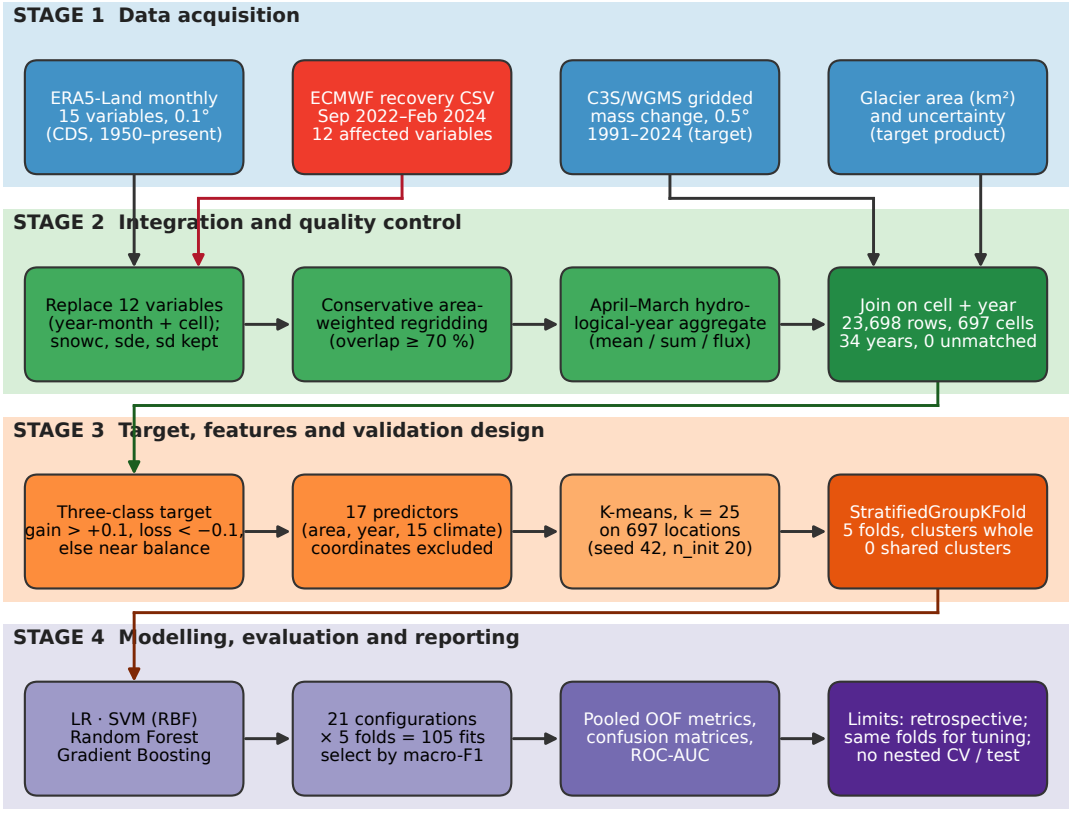
STAGE 1 Data acquisition
ERA5-Land monthly
15 variables, 0.1°
(CDS, 1950–present)
ECMWF recovery CSV
Sep 2022–Feb 2024
12 affected variables
C3S/WGMS gridded
mass change, 0.5°
1991–2024 (target)
Glacier area (km²)
and uncertainty
(target product)
STAGE 2 Integration and quality control
Replace 12 variables
(year-month + cell);
snowc, sde, sd kept
Conservative area-
weighted regridding
(overlap ≥ 70 %)
April–March hydro-
logical-year aggregate
(mean / sum / flux)
Join on cell + year
23,698 rows, 697 cells
34 years, 0 unmatched
STAGE 3 Target, features and validation design
Three-class target
gain > +0.1, loss < −0.1,
else near balance
17 predictors
(area, year, 15 climate)
coordinates excluded
K-means, k = 25
on 697 locations
(seed 42, n_init 20)
StratifiedGroupKFold
5 folds, clusters whole
0 shared clusters
STAGE 4 Modelling, evaluation and reporting
LR · SVM (RBF)
Random Forest
Gradient Boosting
21 configurations
× 5 folds = 105 fits
select by macro-F1
Pooled OOF metrics,
confusion matrices,
ROC-AUC
Limits: retrospective;
same folds for tuning;
no nested CV / test